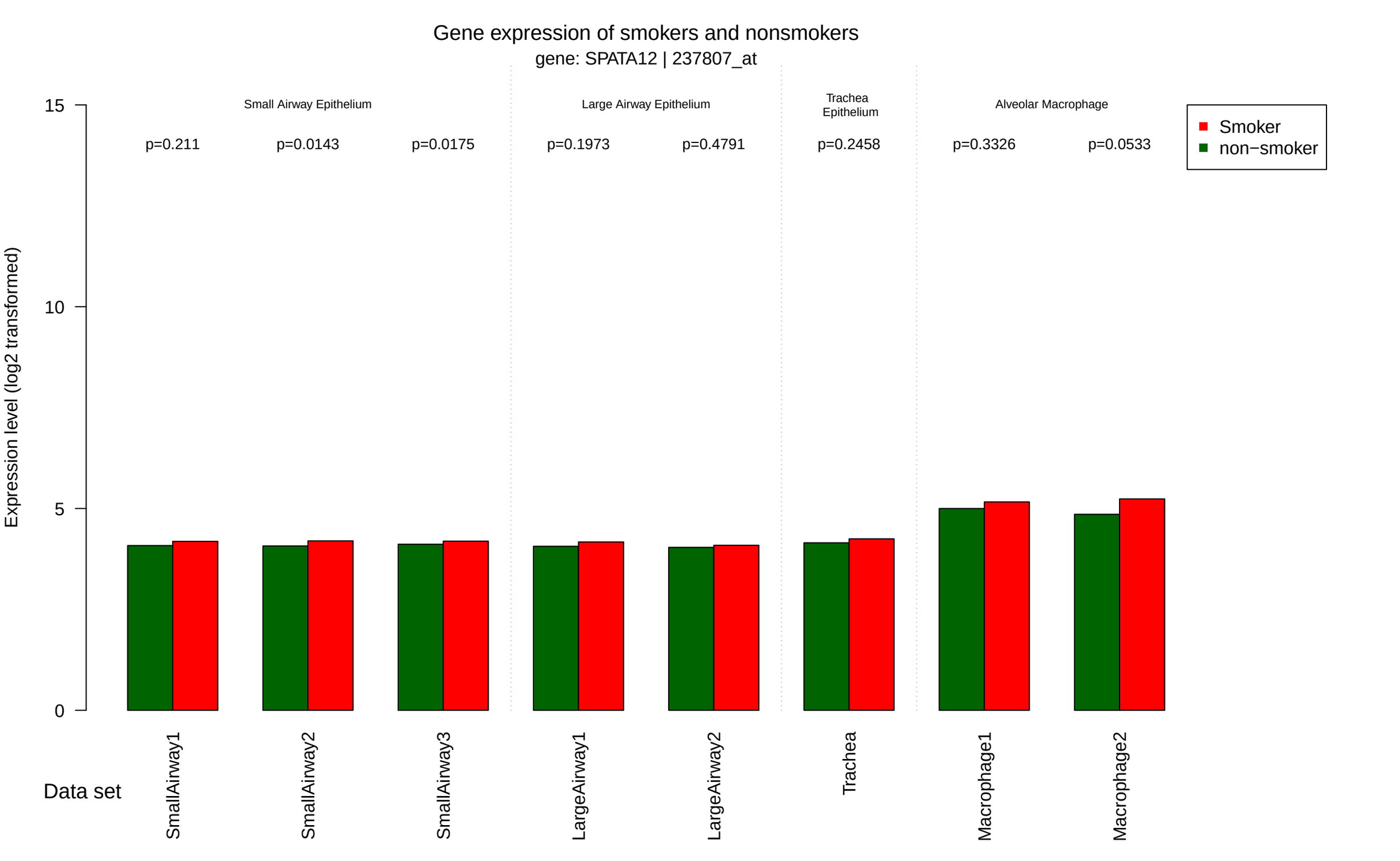
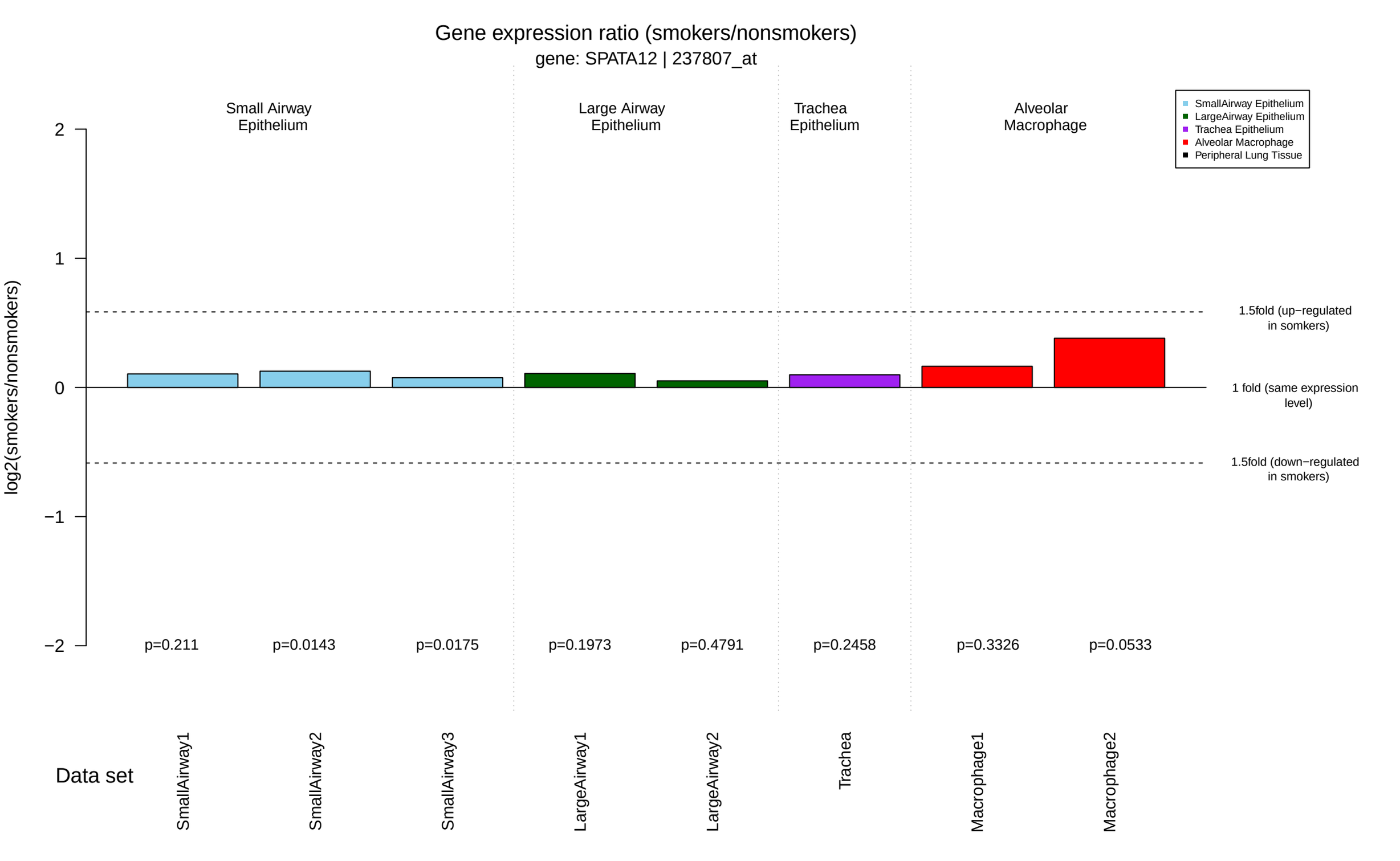
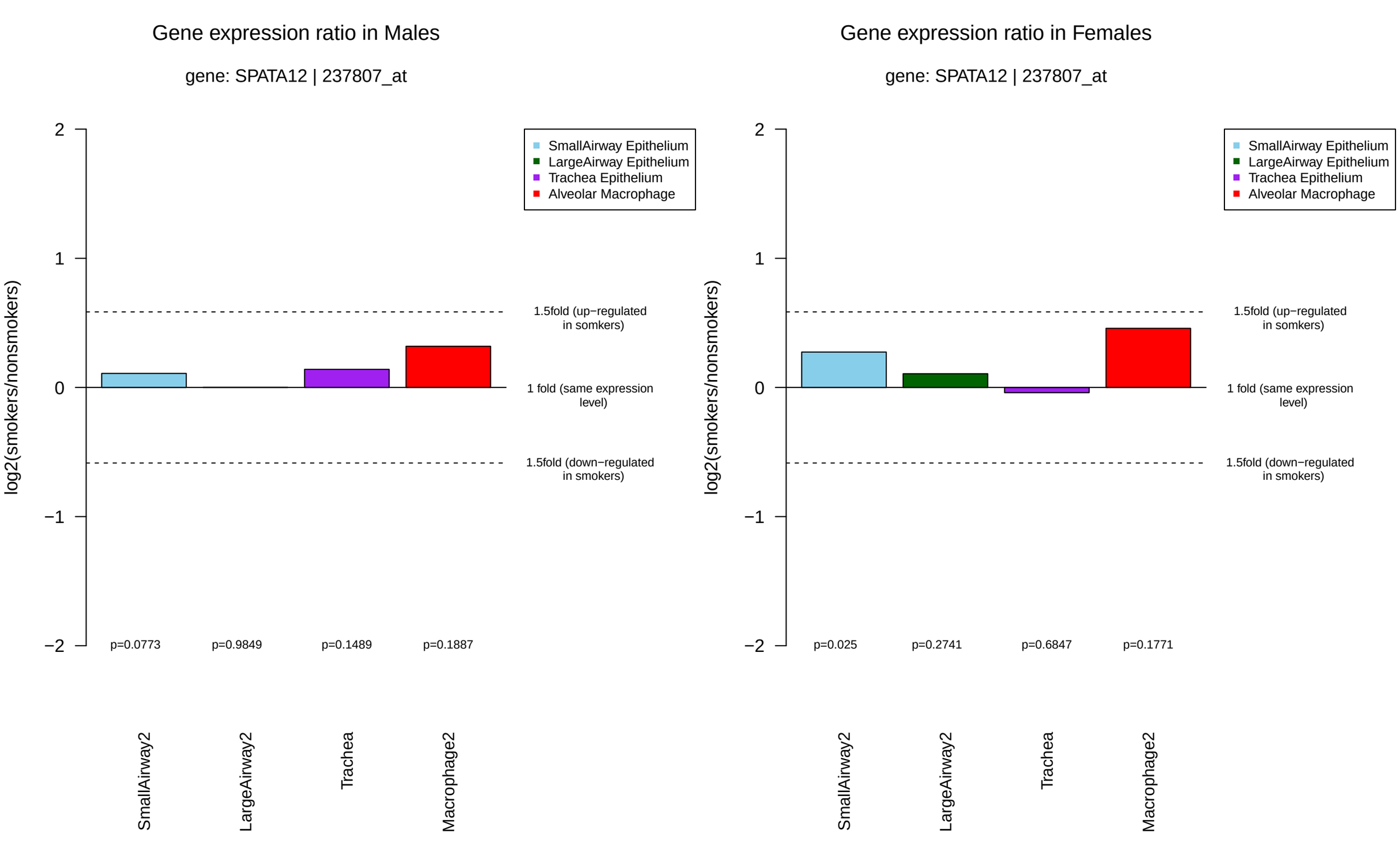
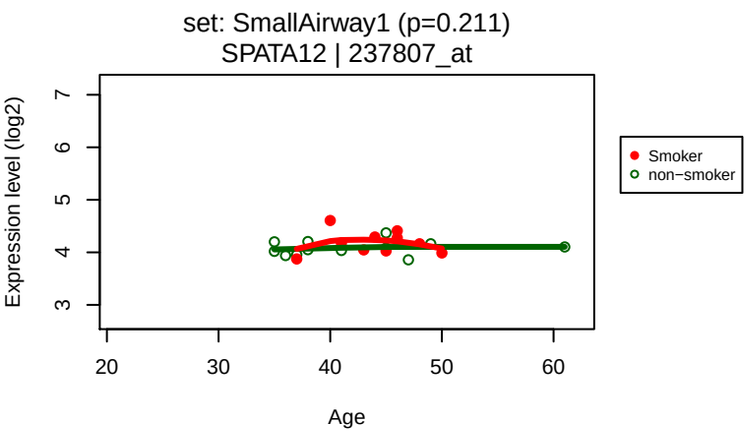
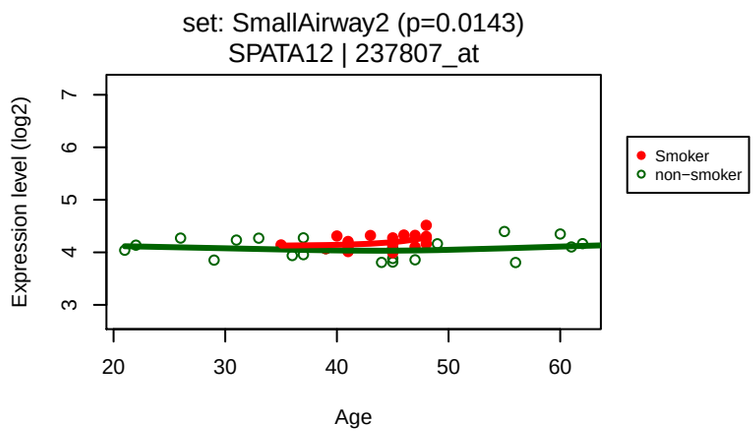
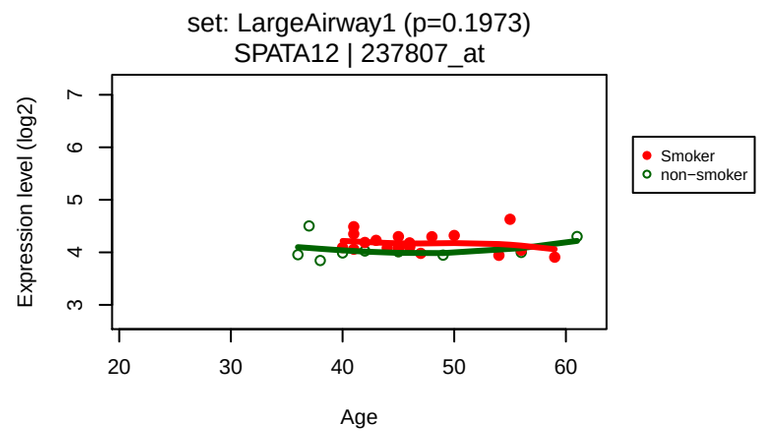

Gene expression of smokers and nonsmokers
gene: SPATA12 | 237807_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.211
p=0.0143
p=0.0175
p=0.1973
p=0.4791
p=0.2458
p=0.3326
p=0.0533
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

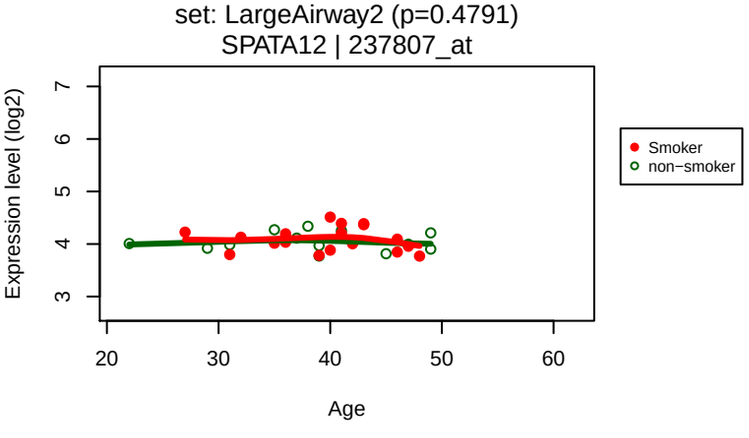

Gene expression ratio (smokers/nonsmokers)
gene: SPATA12 | 237807_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.211
p=0.0143
p=0.0175
p=0.1973
p=0.4791
p=0.2458
p=0.3326
p=0.0533
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

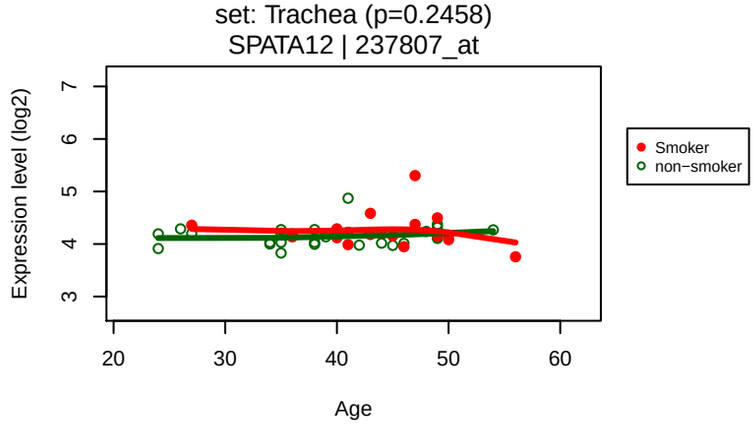

Gene expression ratio in Males
gene: SPATA12 | 237807_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0773
p=0.9849
p=0.1489
p=0.1887
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPATA12 | 237807_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.025
p=0.2741
p=0.6847
p=0.1771
SmallAirway2
LargeAirway2
Trachea
Macrophage2

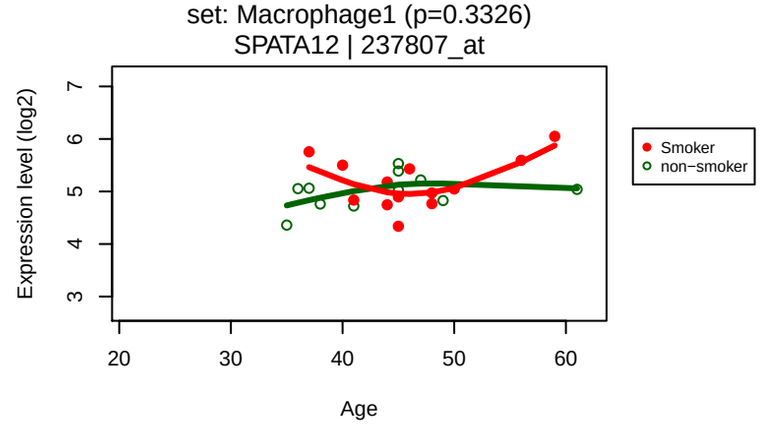

set: SmallAirway1 (p=0.211)
SPATA12 | 237807_at
Expression level (log2)
Age
Smoker
non−smoker

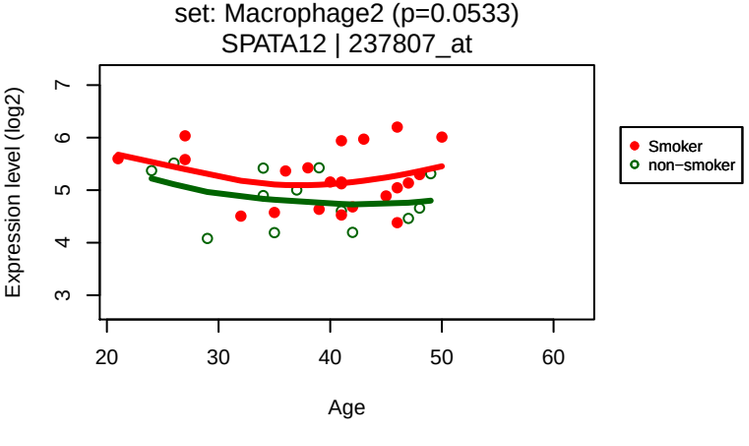

set: SmallAirway2 (p=0.0143)
SPATA12 | 237807_at
Expression level (log2)
Age
Smoker
non−smoker

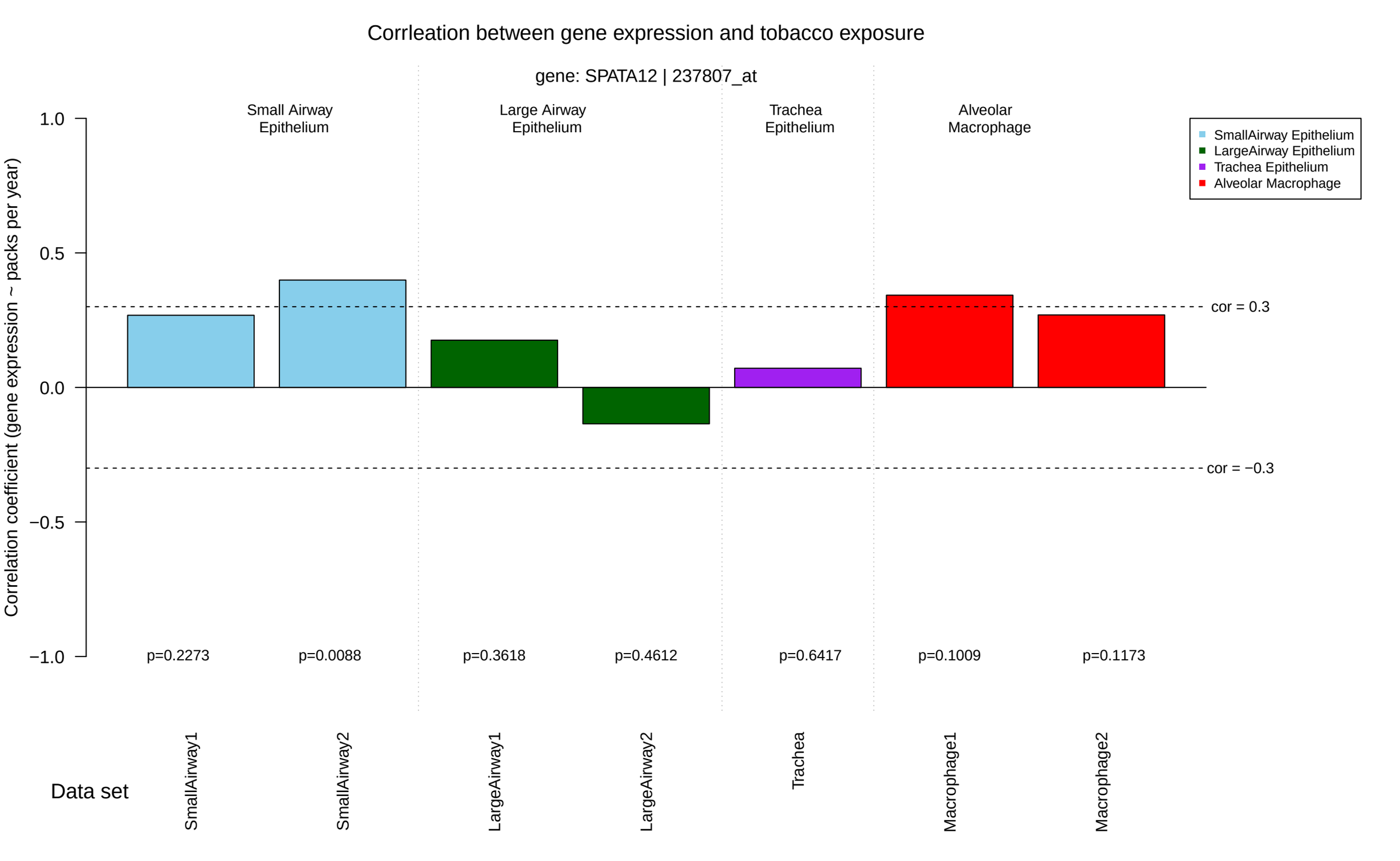

set: LargeAirway1 (p=0.1973)
SPATA12 | 237807_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4791)
SPATA12 | 237807_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2458)
SPATA12 | 237807_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3326)
SPATA12 | 237807_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0533)
SPATA12 | 237807_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPATA12 | 237807_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2273
p=0.0088
p=0.3618
p=0.4612
p=0.6417
p=0.1009
p=0.1173
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2